

MUSTAFA İHSAN AKAGÜNDÜZ

Çankaya, Ankara · +90 546 556 61 10 · akagunduzmustafa00@gmail.com
linkedin.com/in/mustafa-akagunduz · github.com/mustafaakagunduz · mustafa-akagunduz.dev

SUMMARY

Full-stack Software Engineer experienced in building scalable, real-time web applications with React, TypeScript, Python/Django, and NestJS. Hands-on experience with geospatial visualization and real-time rendering of live vehicle/sensor data using CesiumJS and MapLibre. Strong foundations in PostgreSQL/PostGIS, REST APIs, WebSockets, Docker, and CI/CD, with a focus on system reliability and clean, maintainable code.

EDUCATION

Hacettepe University | B.Sc. in Computer Engineering

Ankara, Turkey

EXPERIENCE

Fullstack Developer | Da Vinci Board Game

Oct 2025 – Present

- Design and implement end-to-end features for a high-traffic business management platform using React and NestJS.
- Build real-time data flows with WebSockets, enabling live operational updates across the system.
- Improve performance and scalability by optimizing REST APIs, MongoDB queries, and Redis caching, and implement concurrency-safe workflows ensuring data consistency across integrated systems.
- Integrate third-party marketplace APIs (Shopify, Trendyol, Hepsiburada) to synchronize orders, inventory, and operational workflows.

Software Engineer Intern | OBSS Technology

Jul 2025 – Aug 2025

- Built a real-time aircraft tracking SPA with React, Vite, and CesiumJS, rendering live geospatial data (position, altitude, heading) on an interactive 3D map.
- Handled high-frequency position updates with a focus on performant rendering and smooth visualization of moving entities.
- Designed and implemented the full frontend architecture independently, delivering a production-ready solution.
- Ranked 1st among all interns with the project.

Software Engineer Intern | TÜBİTAK BİLGEM YTE

Jun 2024 – Aug 2024

- Developed a full-stack personnel management system using Spring Boot, Next.js, and PostgreSQL.
- Built backend APIs, relational database schemas, and responsive frontend components in an Agile team.

Software Engineer Intern | TURKCELL İletişim A.Ş.

Aug 2023 – Sept 2023

- Built a backend documentation system to improve developer onboarding; contributed to testing during a legacy CRM migration.

PROJECTS

TRANSYÜK | Logistics Platform

Apr 2025 – Present

- Building a scalable logistics platform connecting carriers, brokers, and shippers, with backend architecture and real-time workflows for supply chain efficiency.

SkillFit | Graduation Project

Oct 2024 – Jun 2025

- Developed a full-stack career recommendation platform (Spring Boot, React, TypeScript, PostgreSQL) with an AI-based recommendation engine; deployed on AWS and Vercel.

Smart Greenhouse IoT System | IoT & Embedded Systems

- Built an IoT-based smart greenhouse monitoring and control system on ESP32 (Wokwi), reading temperature, humidity, and soil-moisture data from simulated sensors.
- Implemented MQTT communication (Mosquitto/HiveMQ) with a Node-RED dashboard for real-time monitoring and remote control of actuators such as cooling and irrigation.

TECHNICAL SKILLS

Languages: Python, TypeScript / JavaScript, Java, SQL

Backend: Django, Django REST Framework, NestJS, Spring Boot

Frontend: React (Vite, Next.js), HTML5, CSS3, responsive UI design

Geospatial: CesiumJS, MapLibre GL JS, PostGIS, GeoJSON, vector & raster data fundamentals

Databases: PostgreSQL (PostGIS), MongoDB, Redis

DevOps & Tools: Docker, Git, CI/CD (GitHub Actions), Linux, AWS, Vercel

Other: REST APIs, WebSockets, GraphQL, JWT Authentication, ORMs, automated testing

EXTRACURRICULAR

Content Creator | YouTube – “Ben Kisaca AKA”

Jun 2024 – Present

- Create content on software engineering and problem-solving, explaining technical topics in a simplified, engaging way.